

Lab AggStates



Agregados en ClickHouse

El **objetivo** de este laboratorio es mostrar la conjunción de mecanismos de *ClickHouse*, fundamentalmente la [familia de motores MergeTree](#), los [estados de agregación](#) y las [vistas materializadas](#) que explican el rendimiento alcanzado por este gestor en el procesamiento de las consultas de agregados

DDL

```
-- DDL tabla
DROP TABLE IF EXISTS etsiinf.Events;
CREATE TABLE etsiinf.Events (
    EventDate DateTime('Europe/Madrid') NOT NULL,
    EventID UInt8 NOT NULL,
    V1 UInt32 NOT NULL,
    V2 UInt32 NOT NULL
)
ENGINE = MergeTree ORDER BY (EventDate, EventID);
```



DML

```
SELECT toStartOfHour(EventDate), EventDate, EventID, V1, V2 FROM
etsiinf.Events;

-- 1ª tanda
INSERT INTO etsiinf.Events VALUES (toDate('2025-04-28 01:03:00',
'Europe/Madrid'), 1, 1, 3);
INSERT INTO etsiinf.Events VALUES (toDate('2025-04-28 01:02:00',
'Europe/Madrid'), 1, 2, 3);
INSERT INTO etsiinf.Events VALUES (toDate('2025-04-28 02:02:00',
'Europe/Madrid'), 2, 3, 2);
INSERT INTO etsiinf.Events VALUES (toDate('2025-04-28 01:03:00',
'Europe/Madrid'), 1, 3, 3);
INSERT INTO etsiinf.Events VALUES (toDate('2025-04-28 02:04:00',
'Europe/Madrid'), 2, 2, 4);

-- 2ª tanda
INSERT INTO etsiinf.Events VALUES (toDate('2025-04-28 03:03:00',
'Europe/Madrid'), 1, 1, 3);
INSERT INTO etsiinf.Events VALUES (toDate('2025-04-28 03:02:00',
'Europe/Madrid'), 1, 2, 3);
INSERT INTO etsiinf.Events VALUES (toDate('2025-04-28 03:02:00',
'Europe/Madrid'), 2, 3, 2);
```



```
INSERT INTO etsiinf.Events VALUES (toDateTime('2025-04-28 03:03:00',
'Europe/Madrid'), 1, 3, 3);
INSERT INTO etsiinf.Events VALUES (toDateTime('2025-04-28 03:04:00',
'Europe/Madrid'), 2, 2, 4);
```

AggregatingMergeTree



Versión con vista materializada

En este caso se crea una vista materializada denominada Events_mv sobre la tabla *Events*. En la vista se definen tres columnas **sV1**, **mV2** y **uV2** que son combinadores **-State** con tipo `AggregateFunction`. Por consiguiente, no es posible consultar estas columnas directamente pues son **estados intermedios de agregación**. En la consulta debe invocarse el combinador **-Merge** correspondiente.

DDL: Vista AggregatingMergeTree

```
-- DDL vista
DROP VIEW etsiinf.Events_mv;
CREATE MATERIALIZED VIEW etsiinf.Events_mv
(
    EventDate DateTime('Europe/Madrid') NOT NULL,
    EventID UInt8,
    sV1 AggregateFunction(sum, UInt32), -- suma de V1s
    mV2 AggregateFunction(avg, UInt32), -- media de V2s
    uV2 AggregateFunction(uniq, UInt32) -- V2s diferentes
)
ENGINE = AggregatingMergeTree() ORDER BY (EventDate, EventID)
AS SELECT
    toStartOfHour(EventDate) AS EventDate,
    EventID,
    sumState(V1) AS sV1,
    avgState(V2) AS mV2,
    uniqState(V2) AS uV2
FROM etsiinf.Events
GROUP BY EventDate, EventID;
```


DML: consulta sobre la vista


```
```sql
-- OJO: combinadores -Merge
SELECT
 toString(EventDate) AS Fecha,
 EventID AS EvID,
 sumMerge(sV1) AS sV1,
 avgMerge(mV2) AS mV2,
 uniqMerge(uV2) AS uV2
FROM etsiinf.Events_mv
GROUP BY EventDate, EventID
ORDER BY EventDate, EventID;
```


```



Resultado

Tanda

Tabla Events

Vista Events_mv

1

| tsStartOfHour(EventDate) | EventDate | EventID | V1 | V2 |
|--------------------------|-------------------------------|---------|----|----|
| 2025-04-28 01:00:00 | 2025-04-28 01:02:00 GMT+02:00 | 1 | 2 | 3 |
| 2025-04-28 01:00:00 | 2025-04-28 01:03:00 GMT+02:00 | 1 | 1 | 3 |
| 2025-04-28 01:00:00 | 2025-04-28 01:03:00 GMT+02:00 | 1 | 3 | 3 |
| 2025-04-28 02:00:00 | 2025-04-28 02:02:00 GMT+02:00 | 2 | 3 | 2 |
| 2025-04-28 02:00:00 | 2025-04-28 02:04:00 GMT+02:00 | 2 | 2 | 4 |

| A7 Fecha | EVID | SV1 | mV2 | W2 |
|---------------------|------|-----|-----|----|
| 2025-04-28 01:00:00 | 1 | 6 | 3 | 1 |
| 2025-04-28 02:00:00 | 2 | 5 | 3 | 2 |

2

| tsStartOfHour(EventDate) | EventDate | EventID | V1 | V2 |
|--------------------------|-------------------------------|---------|----|----|
| 2025-04-28 01:00:00 | 2025-04-28 01:02:00 GMT+02:00 | 1 | 2 | 3 |
| 2025-04-28 01:00:00 | 2025-04-28 01:03:00 GMT+02:00 | 1 | 1 | 3 |
| 2025-04-28 01:00:00 | 2025-04-28 01:03:00 GMT+02:00 | 1 | 3 | 3 |
| 2025-04-28 02:00:00 | 2025-04-28 02:02:00 GMT+02:00 | 2 | 3 | 2 |
| 2025-04-28 02:00:00 | 2025-04-28 02:04:00 GMT+02:00 | 2 | 2 | 4 |
| 2025-04-28 03:00:00 | 2025-04-28 03:02:00 GMT+02:00 | 1 | 2 | 3 |
| 2025-04-28 03:00:00 | 2025-04-28 03:02:00 GMT+02:00 | 2 | 3 | 2 |
| 2025-04-28 03:00:00 | 2025-04-28 03:03:00 GMT+02:00 | 1 | 1 | 3 |
| 2025-04-28 03:00:00 | 2025-04-28 03:03:00 GMT+02:00 | 1 | 3 | 3 |
| 2025-04-28 03:00:00 | 2025-04-28 03:04:00 GMT+02:00 | 2 | 2 | 4 |

| A7 Fecha | EVID | SV1 | mV2 | W2 |
|---------------------|------|-----|-----|----|
| 2025-04-28 01:00:00 | 1 | 6 | 3 | 1 |
| 2025-04-28 02:00:00 | 2 | 5 | 3 | 2 |
| 2025-04-28 03:00:00 | 1 | 6 | 3 | 1 |
| 2025-04-28 03:00:00 | 2 | 5 | 3 | 2 |



Versión con tabla

En este segundo ejemplo se crea una tabla denominada *aggEvents* con motor *AggregatingMergeTree* y las mismas tres columnas *sV1*, *mV2* y *uV2* del ejemplo anterior. En este caso, la vista actualiza directamente sobre la tabla (cláusula *TO*)

DDL: tabla *AggregatingMergeTree* para almacenar los resultados de la vista

```
DROP TABLE IF EXISTS etsiinf.aggEvents;
CREATE TABLE etsiinf.aggEvents
(
    EventDate DateTime('Europe/Madrid') NOT NULL,
    EventID UInt8,
    sV1 AggregateFunction(sum, UInt32), -- suma de V1s
    mV2 AggregateFunction(avg, UInt32), -- media de V2s
    uV2 AggregateFunction(uniq, UInt32) -- V2s diferentes
)
ENGINE = AggregatingMergeTree() ORDER BY (EventDate, EventID);
```

DDL: vista para los estados de las funciones de agregación

```
DROP VIEW etsiinf.aggEvents_mv;
CREATE MATERIALIZED VIEW etsiinf.aggEvents_mv
TO etsiinf.aggEvents
AS SELECT
    toStartOfHour(EventDate) AS EventDate,
    EventID,
    sumState(V1) AS sV1,
    avgState(V2) AS mV2,
    uniqState(V2) AS uV2
FROM etsiinf.Events
GROUP BY EventDate, EventID;
```

DML: consulta directamente sobre la tabla

```
SELECT
    toString(EventDate) AS Fecha,
    EventID AS EvID,
    sumMerge(sV1) AS sV1,
    avgMerge(mV2) AS mV2,
    uniqMerge(uV2) AS uV2
FROM etsiinf.aggEvents
GROUP BY EventDate, EventID
ORDER BY EventDate, EventID;
```



Resultado

Logicamente, es exactamente el mismo que en el caso anterior 👁

Ejercicio



Automatización de acumulados y borrados

Extrapolar el planteamiento del ejercicio anterior a los casos de uso **automatización de acumulados** ([SummingMergeTree](#)) y/o **eliminación de nulos o duplicados** ([ReplacingMergeTree](#), [CoalescingMergeTree](#), [CollapsingMergeTree](#))